Gradient of a line

1

a The line has a gradient of zero.

b The line has an undefined gradient.

c The line has a positive gradient.

d The line has a negative gradient.

2

a i 2 ii 6 iii $\frac{1}{3}$

b i 12 ii 6 iii 2

c i 14 ii 7 iii 2

d i 12 ii 4 iii 3

e i -12 ii 4 iii -3

f i -6 ii 2 iii -3

g i -6 ii 6 iii -1

h i 8 ii 8 iii 1

3 Because a vertical line has no run. So the formula for the gradient $\frac{\text{rise}}{\text{run}}$ will have a denominator of zero, and any fraction with a sero denominator is undefined.

4 0

5 **a** 2 **b** $-\frac{7}{5}$ **c** $\frac{1}{2}$ **d** $-\frac{2}{3}$

Gradient formula

6 $-\frac{3}{2}$

7 No

8

a -2

b As x increases, the value of y decreases.

9 **a** 3 **b** $\frac{3}{5}$ **c** -4 **d** -3

e 0 **f** 0 **g** 1 **h** -1

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10

a $\frac{5}{3}$ **b** $\frac{5}{3}$ **c** Yes

11 $m = 1$

12

a $t = 3$ **b** $t = 15$ **c** $d = 20$ **d** $c = -\frac{12}{7}$

13

a $Q\,(4, -2)$ **b** $R\,(-4, -18)$ **c** 2

14

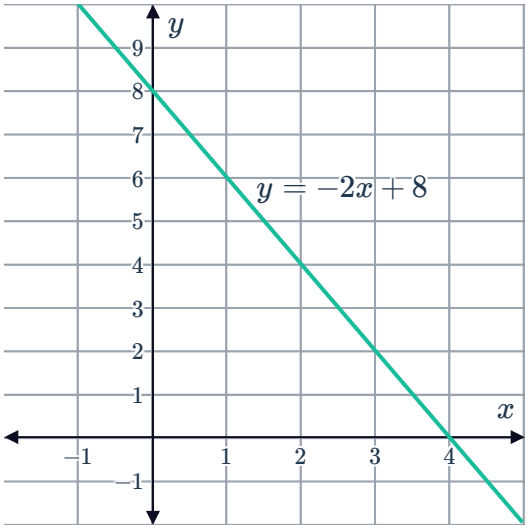
a $y = 27$ **b** 5

15

a

i $y = 8$ **ii** $x = 4$

b



c -2

16

a $y = 1$ **b** $y = 1$ **c**

i 1 **ii** -1

d -1

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Properties of polygons

17

a

i $-\frac{2}{3}$

ii $-\frac{2}{3}$

iii 1

iv 1

b A parallelogram because opposite sides are parallel because they have the same gradient.

18

a -1 **b** 1 **c** -1 **d** Yes

19

a -1 **b** -1 **c** Yes **d** Yes

e Parallelogram

20

a $D\left(\frac{13}{2}, -4\right)$ and $E\left(\frac{1}{2}, -\frac{17}{2}\right)$

b $\frac{3}{4}$

c $\frac{3}{4}$

d Yes

21

a BC

b 15 units^2

Applications

22

a 0.9

b No

23

a -0.72 **b** -0.35 **c** Run A

24 890 m

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